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Gandhian College of Engg. & Tech. BHILAI

324871 (24)

BE (8th Semester)

Examination, Nov.-Dec., 2013

Branch : Electrical, EEE

EHV AC & DC TRANSMISSION (PE-III)

Time Allowed : Three Hours

Maximum Marks : 80

Minimum Pass Marks : 28

Note : (i) Attempt all questions.

(ii) (a) part of each question is compulsory and solve any two parts of remaining 3 parts.

Q. 1. (a) What is Garaetz circuit. 2

(b) Show that in a 3-phase bridge rectifier operating with no-delay and with 60° overlap,

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the direct current is half of the crest value of the line short ckt current on the valve side of the transformer bank. 7

(c) Explain in detail the various kinds of DC links. 7

(d) What are the limitations and advantages of AC and DC transmission. 7

Q. 2. (a) What is FACTS ? 2

(b) Describe briefly various devices used in FACTS system. 1

(c) Explain clearly what do you mean by compensation of line and discuss briefly different methods of compensation.

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(d) Explain with neat diagram the operation of basic TCR.

Q. 3. (a) What is travelling wave ?

(b) Explain the variation of current and voltage on an overhead line when one end of line is : 7

(i) Short-ckted and

(ii) Open ckted and at other end a source of constant emf V is switched in.

(c) A surge of 15 KV magnitude travels along a cable towards its junction with an overhead

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line. The inductance and capacitance of the cable and overhead line are respectively 0.3 mH, 0.4 μF and 1.5 mH, 0.012 μF per km. Find the voltage rise at the junction due to the surge. 7

(d) Show that a travelling wave moves with a velocity of light on the overhead line and its speed is proportional to $\frac{1}{\sqrt{\epsilon_r}}$ on a cable with dielectric material of permittivity. 7

Q. 4. (a) Why reactive power compensation is required in transmission line ? 2

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(b) Discuss briefly different methods of ..... compensation. 7

(c) Explain with diagrams some arrangements where by certain harmonics can be eliminated in TCR circuits. 7

(d) Discuss the problems associated with charging long uncompensated line and suggest the method of overcome this problem. 7

Q. 5. (a) What do you mean by EHV dc system. 2

(b) Explain the problems associated with parallel operation of HVAC & DC system. 7

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(c) Explain the methods of regulation of power flow in an HVDC system.

(d) Explain the desired features of control of EHV dc system. 7

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